Chemical Technology, Control and Management

Manuscript 1663

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ISSN 1815-4840, E-ISSN 2181-1105
Himičeskaâ tehnologiâ. Kontrol' i upravlenie
CHEMICAL TECHNOLOGY. CONTROL AND MANAGEMENT
2025, №2 (122) pp.68-73.
International scientific and technical journal
journal homepage: https://ijctcm.researchcommons.org/journal/
Since 2005
Article history: Received 11 February 2025; Received in revised form 18 April 2025; Accepted 30 April 2025; Available online 27 May 2025

DETECTION OF HTTP FLOOD ATTACKS BASED ON MACHINE LEARNING ALGORITHMS

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Abstract: *This paper analyzes the effectiveness of Random Forest and SVM models for detecting HTTP Flood attacks. Experimental results demonstrate that both models achieve high accuracy. Evaluation was conducted using Precision, Recall, and F1 Score metrics. Additionally, key features of network traffic were extracted through correlation analysis to enable real-time application of the models in attack detection. The findings provide important insights into detecting DDoS attacks using machine learning and improving model performance.*
Keywords: *HTTP Flood, DDoS, Machine Learning, Random Forest, SVM, Overfitting, Precision, Recall, F1 Score, Cybersecurity.*

Annotatsiya*: Ushbu maqolada HTTP Flood hujumlarini aniqlash uchun Random Forest va SVM modellarining samaradorligi tahlil qilingan. Eksperimnet natijalari har ikkala modelning yuqori aniqlikka ega ekanligini ko‘rsatdi. Ularni Precision, Recall va F1 Score metrikalari yordamida baholash amalga oshirildi. Shuningdek, modellarni real vaqtda qo‘llash maqsadida korrelyatsiya tahlili asosida hujumlarni aniqlash uchun tarmoq trafigining asosiy xususiyatlari ajratildi. Tadqiqot natijalari DDoS hujumlarini mashinali o‘rganish yordamida aniqlash va model sifatini oshirish bo‘yicha muhim xulosalar beradi.*
Tayanch so‘zlar: *HTTP Flood, DDoS, Mashina o‘rganish, Random Forest, SVM, Overfitting, Precision, Recall, F1 Score, Kiberxavfsizlik.*

Аннотация: *В работе анализируется эффективность моделей Random Forest и SVM для обнаружения HTTP-флуд-атак. Результаты экспериментов свидетельствуют о том, что обе модели обеспечивают высокую точность. Оценка проводилась с использованием показателей точности, отзыва и F1 Score. Кроме того, с помощью корреляционного анализа были получены ключевые характеристики сетевого трафика, позволяющие применять модели для обнаружения атак в режиме реального времени. Полученные результаты дают важную информацию о том, как обнаруживать DDoS-атаки с помощью машинного обучения и повышать производительность моделей.*
Ключевые слова: *HTTP-флуд, DDoS-атаки, машинное обучение, случайный лес, SVM, переобучение, точность, отзыв, оценка F1, кибербезопасность.*

Introduction

In recent years, the security of internet infrastructure has emerged as a global concern. In particular, Distributed Denial of Service (DDoS) attacks have caused significant damage to various online network resources, including online services and social media platforms [1, 2, 3]. Among the most dangerous forms of DDoS attacks are HTTP Flood attacks, which aim to overwhelm servers by sending excessive requests, thereby disrupting the availability of services. Traditional security mechanisms have proven to be insufficient in detecting and preventing such attacks, leading to the increased adoption of machine learning techniques in this domain.

This study focuses on detecting HTTP Flood attacks using the Random Forest and Support Vector Machine (SVM) models, and analyzing their effectiveness. These algorithms were chosen due to their high performance in processing complex and noisy data, which is characteristic of DDoS traffic.

The research investigates the potential of applying machine learning methods for more accurate classification of HTTP Flood attacks. The evaluation is based on metrics such as precision, recall, and F1 score. Furthermore, the study addresses the problem of overfitting during model training and applies correlation analysis to identify the most critical features for effective attack detection.

The results of this research provide a foundation for enhancing defense strategies against DDoS attacks using machine learning techniques. This work proposes innovative approaches aimed at real-time detection and prevention of cybersecurity threats.

Literature Review and Methods

HTTP Flood attacks are currently among the most common forms of DDoS threats. Numerous researchers have investigated various methods for detecting and preventing such attacks. Studies indicate that modern technologies and techniques play a crucial role in the effective identification of these threats. For example, in [4, 5, 6], researchers utilized network traffic monitoring and anomaly detection techniques to identify DDoS attacks. They argue that early detection of attacks requires careful analysis of traffic flow and fluctuations in the number of HTTP requests.

Moreover, studies conducted in 2017 and 2023 [7,8] have demonstrated high efficiency in detecting attacks using machine learning methods, particularly Random Forest and Support Vector Machine (SVM) algorithms. These algorithms enable effective monitoring of traffic feature variations and improve the ability to distinguish between legitimate and malicious requests.

The Random Forest algorithm is capable of handling a wide variety of features and automatically identifying feature importance, which is critical for extracting key parameters from network traffic. Its resistance to overfitting and ability to handle large datasets make it a strong candidate for accurate and reliable attack analysis.

SVM, on the other hand, is a powerful algorithm for DDoS attack detection, especially when the data contains numerous features. Its use of kernel functions enables the discovery of complex, nonlinear relationships within the data that are essential for identifying attack-specific signatures and ensuring timely detection of threats.

Some researchers have also proposed statistical approaches for analyzing attacks. For instance, A. Kolesnikov [17] employed Poisson and Cauchy distributions to analyze the volume and frequency of HTTP requests. This method highlights the ability to differentiate between normal and anomalous traffic patterns using statistical modeling.

Thus, previous studies suggest that combining statistical and machine learning techniques provides a complementary and effective approach to HTTP Flood attack detection.

However, the ever-evolving nature of HTTP Flood attack scenarios, the unpredictability of attack targets, the limited real-time applicability of existing machine learning algorithms, and the high computational cost of model training underscore the relevance of this study. The current research aims to address these challenges by evaluating the performance and practical deployment of detection models in real-time environments.

Model Construction for HTTP Flood Attack Detection

In this study, we employed a series of modern approaches to detect and classify HTTP Flood attacks. The model construction process consists of the following stages:

1) Data collection and analysis;
2) Data preprocessing;
3) Machine learning and classification.

HTTP traffic data was collected and subjected to an initial analysis. In this stage, key traffic features—such as request frequency, size, and timing—were examined.

The data was cleaned and transformed into a consistent format. Irrelevant or incorrect entries in the traffic data were removed, and the remaining features were standardized.

To distinguish between normal and malicious requests in the traffic, we utilized the Random Forest and Support Vector Machine (SVM) algorithms. These algorithms effectively track variations in traffic characteristics and demonstrate high performance in separating benign and malicious traffic.

To detect anomalies in traffic behavior, Poisson and Cauchy distributions [24, 25] were applied. These statistical approaches are used to identify deviations from expected (normal) traffic patterns.

For attack detection, threshold-based detection and entropy-based analysis methods were also used to monitor anomalies and differentiate them from legitimate changes in traffic patterns.

The combination of these methods enabled effective detection of HTTP Flood attacks. The next section presents the results and analysis of the proposed model.

Analysis of Results

The models developed based on the proposed algorithms were trained and analyzed using a dataset [19] consisting of 852,585 entries and 9 attributes. The dataset includes various network-related features, which are composed of the following fields:

- Highest Layer – The highest protocol layer of the packet (e.g., HTTP, ARP, UDP);
- Transport Layer – The transport protocol (e.g., TCP, UDP);
- Source IP and Destination IP – The sender and receiver IP addresses;
- Source Port and Destination Port – The port numbers the packet is sent from and to;
- Packet Length – The size of the packet in bytes;
- Packets/Time – The number of packets received in a given time interval;
- Target – The labeled result for classification: normal or attack traffic.

A correlation analysis of these features was conducted to measure the relationships between variables. Using the corr() function, the dependencies between various attributes of the network traffic were identified, helping to determine which features are most relevant for detecting attacks.

The correlation coefficient is mathematically calculated as follows:

$$r_{xy} = \frac{\sum(x_i - \breve{x})(y_i - \breve{y})}{\sqrt{\sum(x_i - \breve{x})^2 \sum(y_i - \breve{y})^2}}$$

where, x and y are the values of two variables; $\breve{x}$ and $\breve{y}$ are their respective means; r_{xy} expresses the degree of correlation between the two variables.

Figure 1 shows the correlation analysis results of the dataset features.

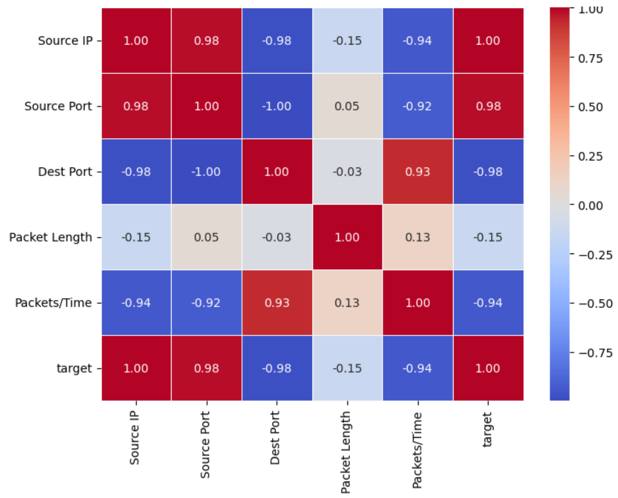

Fig. 1. ***Correlation Analysis of Dataset Features.***

The correlation values from this analysis indicate the following:

– or -1.0 – Strong correlation. A value of 1 means both variables increase or decrease together, while -1 means one increases as the other decreases;

– 0.5 or -0.5 – Moderate correlation, indicating some relationship, but not a strong one;
– 0 or near 0 – No or very weak correlation between the variables.

The analysis revealed that some features play a crucial role in attack detection. In particular, the correlation between Packets/Time and Packet Length was found to be significant. By using corr(), the most important features for classification can be identified and used to optimize the model.

For the purpose of detecting HTTP Flood attacks, a subset of 172,852 HTTP traffic records was extracted and classified as either normal or malicious. The following key features were identified for the classification model:

– HTTP request frequency – Number of requests over a certain time interval;
– Packet size and port activity – Analysis of abnormally high or low packet sizes or port usage;
– IP address and request distribution – Evaluation of excessive requests from a single source.

Based on these features, the performance results of the classification models are presented in Tab.1.

Table 1.

Performance of Classification Models

Model	Precision	Recall	F1-score	Accuracy	Training Time (seconds)
Random Forest	96.4%	95.8%	96.1%	96.3%	1.23s
SVM (Support Vector Machine)	93.7%	92.5%	93.1%	93.6%	2.89s
Poisson Distribution	89.2%	88.5%	88.8%	89.1%	0.98s
Cauchy Distribution	85.6%	84.9%	85.2%	85.5%	1.12s

We now examine the obtained results based on the following performance indicators: Precision, Recall, F1-score, Accuracy, and Execution Time (seconds):

– Precision – Indicates the proportion of correctly predicted attacks among all instances labeled as attacks by the model. A high precision value suggests a low false positive rate;
– Recall – Represents the proportion of actual attacks that were correctly identified by the model. High recall indicates the model's ability to capture most attack instances;
– F1-score – A metric that balances Precision and Recall. If there is a significant difference between Precision and Recall, the F1-score provides a harmonic mean to reflect their trade-off;
– Accuracy – The percentage of total correctly classified instances across all classes. It provides an overall assessment of the model's performance;
– Execution Time (seconds) – Reflects the computational efficiency of the model, indicating how long it takes to process the data and produce results.

Interpretation of Findings:

– The Random Forest model achieved the highest accuracy (96.3%), demonstrating its effectiveness in distinguishing between normal and malicious traffic;
– The SVM model also produced solid results (93.6%), though it required slightly more processing time;
– The statistical approaches (based on Poisson and Cauchy distributions) yielded noticeably lower accuracy rates, indicating that they may not be sufficiently reliable for real-time attack detection.

The analysis confirms that models built using machine learning algorithms are effective in detecting HTTP flood attacks. In particular, Random Forest and SVM algorithms achieved high accuracy and are well-suited for real-time deployment.

Conclusion

The results of this study demonstrate that machine learning-based models, particularly Random Forest and SVM, are more effective than statistical approaches for detecting HTTP flood attacks. The correlation analysis revealed the significance of features such as Packets/Time and Packet Length in the dataset, which enhances both the detection accuracy and speed of the models.

Among the evaluated models, Random Forest achieved the highest performance metrics, with a precision of 96.4%, recall of 95.8%, F1-score of 96.1%, and an overall accuracy of 96.3%, while requiring only 1.23 seconds for training. This indicates its high efficiency and suitability for real-time application.

The SVM model also yielded solid results, with a precision of 93.7%, recall of 92.5%, and a training time of 2.89 seconds. These findings support the effectiveness of the Random Forest algorithm in constructing a robust detection model for HTTP flood attacks.

For future work, this research can be expanded by: Analyzing large-scale network traffic to detect various types of DDoS attacks, Identifying attack-specific informative features, Evaluating the interdependencies between traffic attributes according to the attack type, and integrating additional relevant data sources to improve detection accuracy and robustness.

References:

1. *"O'zbekiston Respublikasi kiberxavfsizligi - 2023 yil hisoboti"*. https://csec.uz/uz/news/maqolalar/o-zbekiston-respublikasi-kiberxavfsizligi-2023-yil-hisoboti/
2. *Aktualnye kiberugrozy: III kvartal 2024 [Current Cyber Threats: Q3 2024]*. https://www.ptsecurity.com/ru-ru/research/analytics/aktualnye-kiberugrozy-iii-kvartal-2024-goda/#id1
3. *Novye otchety po kiberatakam i sposobam ikh predotvrashcheniya [New reports on cyberattacks and methods of their prevention]*. https://www.infowatch.ru/analytics/daydzhesty-i-obzory/novyye-otchety-po-kiberatakam-i-sposobam-ikh-predotvrascheniya
4. *Significant Cyber Incidents*. https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents
5. Abbasi, M., Shahraki, A., Taherkordi, A. (2021). Deep learning for network traffic monitoring and analysis (NTMA): A survey. *Computer Communications*, 170, 19-41. https://doi.org/10.1016/j.comcom.2021.01.021.
6. Santoso, N. A., Lutfayza, R., Nughroho, B. I., Gunawan, G. (2024). Anomaly detection in network security systems using machine learning. *Journal of Intelligent Decision Support System (IDSS)*, 7(2), 113-120. https://doi.org/10.35335/idss.v7i2.238.
7. Voruganti, N. K., Raju, S. K. (2023). Network Anomaly Detection Using a Random Forest Classifier. *In Information and Knowledge Systems*. https://doi.org/10.52305/XUNI4482.
8. Lei, Y. (2017). Network anomaly traffic detection algorithm based on SVM. *In 2017 International Conference on Robots & Intelligent System (ICRIS)*. https://doi.org/10.1109/ICRIS.2017.61.
9. Liu, J., Liu, F., Ansari, N. (2014). Monitoring and analyzing big traffic data of a large-scale cellular network with Hadoop. *IEEE Network*, 28(4), 32-39.
10. Sivarajah, U., Kamal, M. M., Irani, Z., Weerakkody, V. (2017). Critical analysis of big data challenges and analytical methods. *Journal of Business Research*, 70, 263-286.
11. Zhou, D., Yan, Z., Fu, Y., Yao, Z. (2018). A survey on network data collection. *Journal of Network and Computer Applications*, 116, 9-23.
12. Lee, S., Levanti, K., Kim, H. S. (2014). Network monitoring: Present and future. *Computer Networks*, 65, 84-98.
13. Verma, S., Kawamoto, Y., Fadlullah, Z. M., Nishiyama, H., Kato, N. (2017). A survey on network methodologies for real-time analytics of massive IoT data and open research issues. *IEEE Communications Surveys & Tutorials*, 19(3), 1457-1477.
14. Lotfollahi, M., Siavoshani, M. J., Zade, R.S.H., Saberian, M. (2020). Deep packet: A novel approach for encrypted traffic classification using deep learning. *Soft Computing*, 24(3), 1999-2012.
15. Rahmatov, F. A., Xolmuminov, O. T. (2024). Tarmoq elektron resurslariga DDos hujumlari tahlili. *Raqamli Transformatsiya va Sun'iy Intellekt*, 2(2), 133-137.
16. Raxmatov, F. A. (2024). Veb-ilovalarga tahdidlar va himoya qilishning mavjud usullari tahlili. *Management, Marketing and Finance*, 1(2), 96-99.
17. Kolesnikov, A. V. (2015). *Modelirovaniye setevogo trafika i algoritmy bor'by s peregruzkami na osnove metodov nelineynoy dinamiki i kratkosrochnogo prognozirovaniya vremennykh ryadov [Modeling of Network Traffic and Congestion Control Algorithms Based on Nonlinear Dynamics Methods and Short-Term Time Series Forecasting]*. Dissertatsiya na soiskaniye uchenoy stepeni kandidata tekhnicheskikh nauk, Moskva. 144 p. (in Russina).
18. Htet, P.P., Thanda, S. (2019). Network traffic anomaly detection based on Apache Spark. *In 2019 International Conference on Advanced Information Technologies (ICAIT)*. *IEEE*. 222-226
19. https://www.kaggle.com/datasets/oktayrdeki/ddos-traffic-dataset
20. Guthula, S., Battula, N., Beltiukov, R., Guo, W., Gupta, A. (2023). NetFound: Foundation model for network security. arXiv preprint arXiv:2310.17025. https://arxiv.org/abs/2310.17025
21. Alazab, M. (2022). A discrete time-varying greywolf IoT botnet detection system. *Computer Communications*, 192, 405-416.

22. Lo, W. W., Layeghy, S., Sarhan, M., Gallagher, M., Portmann, M. (2022). E-GraphSAGE: A Graph Neural Network based Intrusion Detection System for IoT. *In NOMS 2022 – IEEE/IFIP Network Operations and Management Symposium*, Budapest, Hungary, 25-29 April. 1-9.
23. Aouedi, O. (2022). Machine learning-enabled network traffic analysis (Doctoral dissertation, Nantes Université). https://theses.hal.science/tel-03966012v2.
24. Alwhbi, I.A., Zou, C.C., Alharbi, R. N. (2024). Encrypted network traffic analysis and classification utilizing machine learning. *Sensors*, 24(11), 3509. https://doi.org/10.3390/s24113509.
25. Abbasi, M., Shahraki, A., Taherkordi, A. (2021). Deep learning for network traffic monitoring and analysis (NTMA): A survey. *Computer Communications*, 170, 19-41. https://doi.org/10.1016/j.comcom.2021.01.021.
26. *Raspredelenie Puassona [Poisson distribution]*. https://ru.wikipedia.org/wiki/Распределение_Пуассона
27. *Raspredelenie Koshi [Cauchy distribution]*. https://ru.wikipedia.org/wiki/Распределение_Коши
28. Hänsch, R. (2025). Handbook of Random Forests: Theory and Applications for Remote Sensing. 300 p. https://doi.org/10.1142/10552